IEEE SSIT/TEMS Tunisia Section Joint Chapter

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IEEE SIGHT Tunisia Section Special Group

T S Y P 1 3 -T E C H N I C A L C H A L L E N G E

P2E: Power to Empower

RemindMeBox:

Smart Pill Management and Elderly Safety System

"Empowering Independent Aging for Tunisia's Elderly"

Special thanks to all our members for their contributions to this project

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Abstract

This report presents RemindMeBox, a multidisciplinary smart pill management and safety system developed to address critical healthcare gaps exacerbated by the global polycrisis. Millions of individuals, particularly in underserved or crisis-affected communities, struggle to maintain safe and consistent medication routines due to fragmented healthcare access, caregiver overload, displacement, stress, and complex treatment schedules.

RemindMeBox is designed to reduce the workload of caregivers and healthcare professionals by integrating automated pill dispensing, real-time IoT-enabled monitoring, adaptive alerts, and an intuitive user interface, also, AI-driven assistance enables the personalization of medication routines to meet diverse individual needs.

Its combination of automation, real-time monitoring, and adaptive support reduces human error, promotes adherence, and enhances overall medication safety. The system's scalable and adaptable design makes it a resilient solution for diverse and crisis-affected settings, contributing to equitable and sustainable healthcare outcomes.

Key words:
smart pill box; medication adherence; elderly safety; IoT devices; automated dispensing; caregiver monitoring; mobile health application; AI assistant; real-time alerts; remote health monitoring; healthcare innovation; medication compliance; digital health; medication management systems ; polycrisis

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Chapter 1: Problem statement and Proposed solution

Introduction

This chapter introduces the RemindMeBox project, which addresses the critical challenge of medication adherence and safety within the context of today's global polycrisis. The solution proposes an innovative smart pill management system that leverages IoT technology, a dedicated mobile caregiver application, and integrated AI-driven support to create a reliable and accessible daily medication routine for vulnerable populations.

1.1 Project Background And Motivation

In crisis-affected areas and unstable environments, up to **70–80%** of individuals face difficulties maintaining consistent medication routines, due to disrupted access, limited caregiver support, and stress-related factors.
The development of RemindMeBox is driven by the urgent need for an innovative, adaptable, and technology-based solution capable of delivering timely reminders, automating dose dispensing, and connecting patients with caregivers and healthcare professionals in a user-friendly and resilient way.

1.2 Problem statement

Vulnerable individuals, particularly older adults, encounter heightened health risks in crisis-affected settings, where safe medication management is often disrupted by multiple, interconnected challenges:

- **Lack of Real-Time Monitoring:** In emergency shelters, community centers, or unstable healthcare settings, staff cannot continuously track adherence, resulting in preventable complications.
- **Complex Treatment Regimens:** Among some refugees with non-communicable diseases, only 59.5% take their hypertension medication and 67% adhere to their dia-

betes regimen.

- **Cognitive and Memory Challenges:** Over one-third of displaced adults managing chronic disease report moderate-to-severe mental health symptoms (e.g., anxiety, PTSD), which are strongly associated with suboptimal medication adherence.

These factors highlight the urgent need for solutions that ensure timely medication, support caregivers, and mitigate health risks in unstable environments.

1.3 Proposed solution : RemindMeBox

RemindMeBox is a smart medication management system designed to empower vulnerable populations and assist healthcare providers in ensuring timely and accurate medication dispensing, especially in hospitals, chronic care settings, or crisis-affected areas where monitoring every patient is challenging.

The device organizes and dispenses medications according to personalized schedules, which are programmed by a healthcare professional or designated caregiver (Super User) using the connected mobile application. Once set up, the ESP32 Smart Box automatically dispenses pills, tracks inventory, and reports adherence. The mobile platform allows both caregivers and healthcare providers to monitor medication routines, receive real-time alerts for missed doses or emergencies, and respond promptly.

The solution ensures consistent medication adherence for eldery and people with chronic diseases, even in socially fragile or crisis-affected environments, improving health outcomes and reducing preventable complications through combining automated dispensing, personalized scheduling, and data-driven feedback.

1.4 RemindMeBox's Mission, Vision, and Values

Our Mission is to empower vulnerable individuals and support healthcare providers by simplifying the management of medication dispensing, ensuring patients receive the right doses at the right times, whether in hospitals, chronic care settings, or crisis-affected areas where monitoring every patient is challenging.

Our vision is to revolutionize medication management worldwide, enabling seamless, reliable, and automated care for all patients, whether in hospitals, remote regions, or crisis-affected areas while empowering healthcare providers to focus on complex care and im-

proving overall health outcomes over the long term.

This project is guided by a set of core values that shape every aspect of RemindMeBox's design, implementation, and user experience. These values ensure that our solution is not only effective, but also ethical, inclusive, and sustainable.

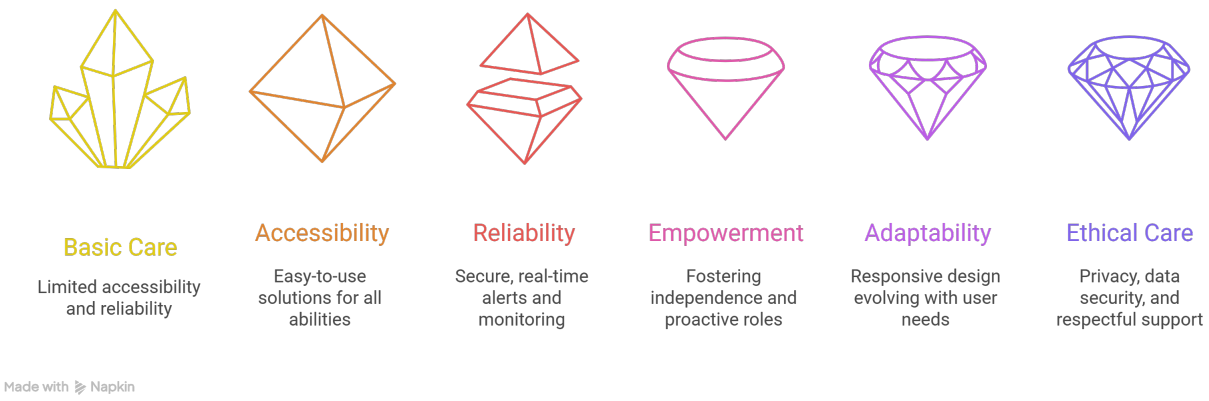


Figure 1.1: RemindMeBox Values

Conclusion

In conclusion, RemindMeBox addresses the challenges of medication management in vulnerable and crisis-affected populations by combining smart dispensing, caregiver-programmed schedules, real-time monitoring, and AI-driven support. The system ensures timely medication, reduces health risks, and empowers healthcare providers to manage patients more effectively, offering a practical and resilient solution for improving adherence and overall care.

Chapter 2: Core Features and Functional Architecture

Introduction

This chapter presents the functional architecture of **RemindMeBox**, illustrating how the smart IoT device, mobile application, and cloud backend interact to deliver a reliable and user-centered medication management experience. The system combines automated dispensing, real-time monitoring, secure data exchange, precise location tracking, and AI-powered assistance to support both elderly users and caregivers.

2.1 System Overview

RemindMeBox operates through a well-organized ecosystem composed of four complementary technological layers.

ESP32 Smart Box

At the core of the system, the ESP32-based smart dispenser executes on-device operations including servo-driven compartment rotation, pill retrieval verification using integrated sensors, stock monitoring, and continuous GPS tracking. After initial pairing via Bluetooth, the device maintains stable communication through **WiFi** to synchronize all operational data with the cloud.

Flutter Mobile Application

The mobile application provides registration, configuration, schedule creation, alert visualization, GPS tracking, and AI-assisted interaction. Two access roles are supported: the **Super User**, responsible for initial configuration and medical setup, and the **Simple User**, dedicated to daily supervision and alerts monitoring.

Firebase Backend

Firebase functions as the central data hub, managing authentication, medication schedules, compartment information, device states, and pill quantities. Through **Firebase**



Figure 2.1: Mobile app design

Cloud Messaging, the system delivers time-sensitive alerts such as missed doses, low stock warnings, or emergency notifications.

External Services

External services enhance system capabilities: the **Gemini AI API** provides conversational and medical assistance, while **OpenStreetMap** combined with the Overpass API ensures precise geolocation and automated detection of nearby hospitals.

2.2 Communication Flow

The communication structure of RemindMeBox is designed for reliability and transparency.

During the initial configuration phase, the mobile application creates a **Bluetooth connection** with the ESP32 to transfer WiFi credentials and the number of pill compartments. Once connected to the internet, the ESP32 switches to **WiFi communication** and syn-

chronizes continuously with Firebase.

All medication schedules and device parameters defined in the mobile app are stored in Firebase, where the ESP32 retrieves them according to its internal synchronization loop. In return, the smart box regularly updates the cloud with stock changes, GPS coordinates, dispensing reports, and emergency events.

The mobile application queries Firebase to display real-time system information, while notification services guarantee the immediate delivery of **missed-dose alerts**, **critical stock warnings**, and **emergency signals**.

2.3 User Flow

The user journey begins with secure authentication via Firebase. Each user selects one of the two roles: **Super User** for configuration or **Simple User** for monitoring. After authentication, the Super User initiates the device onboarding process.

The application detects the ESP32 via Bluetooth, establishes a secure connection, and transmits WiFi credentials together with structural information such as the number of compartments.

Once the smart box connects to the internet, the Super User defines the medications contained in each compartment by specifying the name, quantity, and complementary notes. This information is stored in Firebase and immediately synchronized with the ESP32.

The next step is the creation of personalized medication schedules. Each schedule defines the relevant compartments, the number of pills to dispense, and the exact administration time. Synchronization mechanisms ensure that the ESP32 always maintains an updated list of tasks.

From that point forward, all device interactions are fully remote. Stock variations, GPS data, confirmation of intake, and emergency alerts are periodically synchronized with the cloud and displayed instantly in the mobile application. Caregivers remain informed without requiring physical access to the device.

2.4 Automatic Dispensing and Pill Tracking

The dispensing logic is executed autonomously by the ESP32 based on the updated schedule. When a dose is due, the device triggers the appropriate servo motor to rotate the

compartment and release the required number of pills. Integrated sensors verify whether the pills were retrieved by the user.

If the medication remains uncollected after a configurable delay, the system classifies the event as a **missed dose** and immediately sends notifications to caregivers.

Stock levels are updated automatically after every dispensing event. When the remaining quantity reaches a critical threshold, the system triggers a **low-stock alert** to prevent treatment interruption. Potential hardware anomalies such as servo failures or sensor malfunctions are detected and reported as **technical warnings**.

2.5 GPS Tracking and Safety Boundaries

In addition to medication management, RemindMeBox integrates advanced safety features. The ESP32 periodically records its **GPS coordinates** and transfers them to Firebase. The mobile application interprets this data and displays the user's real-time location on an **OpenStreetMap** interface.

A customizable geofencing radius can be configured by the Super User. When the user exits this area, the system generates an immediate **boundary-breach alert**, indicating potential disorientation or wandering.

Using the Overpass API, the application also identifies nearby hospitals and medical centers, offering rapid guidance in emergency situations.

2.6 AI Chatbot Assistance

The system includes an intelligent assistant powered by the **Gemini API**, accessible from the mobile application. It provides medication explanations, personalized reminders, general health advice, and first-aid guidance during emergencies. This AI component elevates RemindMeBox from a dispensing device to a comprehensive digital health companion.

Conclusion

The functional architecture of RemindMeBox harmoniously combines IoT automation, cloud computing, geolocation monitoring, and AI-driven support. The integration of the ESP32 smart dispenser, the Flutter mobile application, Firebase services, and external APIs creates a robust and user-centered ecosystem. This architecture forms the technological foundation for the device's technical specifications and practical implementation.

Chapter 3: Technical Specifications and Sustainability

Introduction

This Chapter presents the translation of RemindMeBox's conceptual design into a practical, deployable system. It outlines core technical specifications for hardware and software, details power and data management, and introduces sustainability considerations underpinning reliable operation and long-term impact.

3.1 Technical Specifications

This section presents the system's main technical aspects, including its hardware and software architectures, mechanical and electrical designs, and the data management and security mechanisms.

3.1.1 Hardware Architecture

This part describes the system's hardware architecture, detailing the components used, the mechanical structure, and the electrical design.

Component and Budget

The total budget for all the listed components for RemindMeBox is 384.9dt with 82% ROI (return on investment)

Component	Type	Quantity	Unit Price (DT)	Description / Function
Pill Storage Compartment	Mechanical	1	26.80	Durable plastic, modular design for pill storage
Rotor Mechanism	Mechanical	1	20.10	3D printed PLA+ rotor for pill selection and dispensing

Component	Type	Quantity	Unit Price (DT)	Description / Function
Slider Assembly	Mechanical	1	13.40	Guides and slopes ensure pill movement to tray
Main Enclosure	Mechanical	1	40.20	Rigid plastic housing for full assembly
DC Geared Motor (GA12-N2012V 300RPM)	Electrical	3	14	Drives mechanical parts (pill wheel, doors, etc.)
ESP32-WROVER Microcontroller	Electrical	1	34	Central processing, with Wi-Fi/Bluetooth
OLED Display	Electrical	1	17	Shows system status, time, and alerts
Presence Sensor (IR)	Electrical	1	3.8	Detects pill dispensing/retrieval
LED Indicator	Electrical	1	2.3	Shows visual alerts
Push Button	Electrical	1	0.5	User/SOS input
Audio/Buzzer Module	Electrical	1	5	Sound alerts for notifications
12V Battery Turnigy 2200mAh 3S 30C	Electrical	1	140	Main power supply
Power Regulation Module (12V→5V/3.3V)	Electrical	1	14	Voltage conversion for safe operation
Vacuum pump 12V KYK50BPM	Electrical	1	89	Used for suction or pill handling mechanism
Pump Tubing (Ø 4mm)	Mechanical	1	6.25	Connects pump to compartments

Component	Type	Quantity	Unit Price (DT)	Description / Function

Table 3.1: System Components and Budget

Mecanical Design

The mechanical design of RemindMeBox prioritizes precision, durability, and safety to ensure reliable medication delivery. A rigid plastic enclosure houses modular pill compartments that are tamper-resistant and easy to refill. A stepper motor–driven rotor system precisely controls pill dispensing, assisted by smooth slider mechanisms and locking covers for security. Robust fasteners and bearings enhance longevity and reduce wear, while all parts allow easy maintenance and cleaning. This design ensures a secure, durable, and user-friendly device supporting consistent medication adherence.

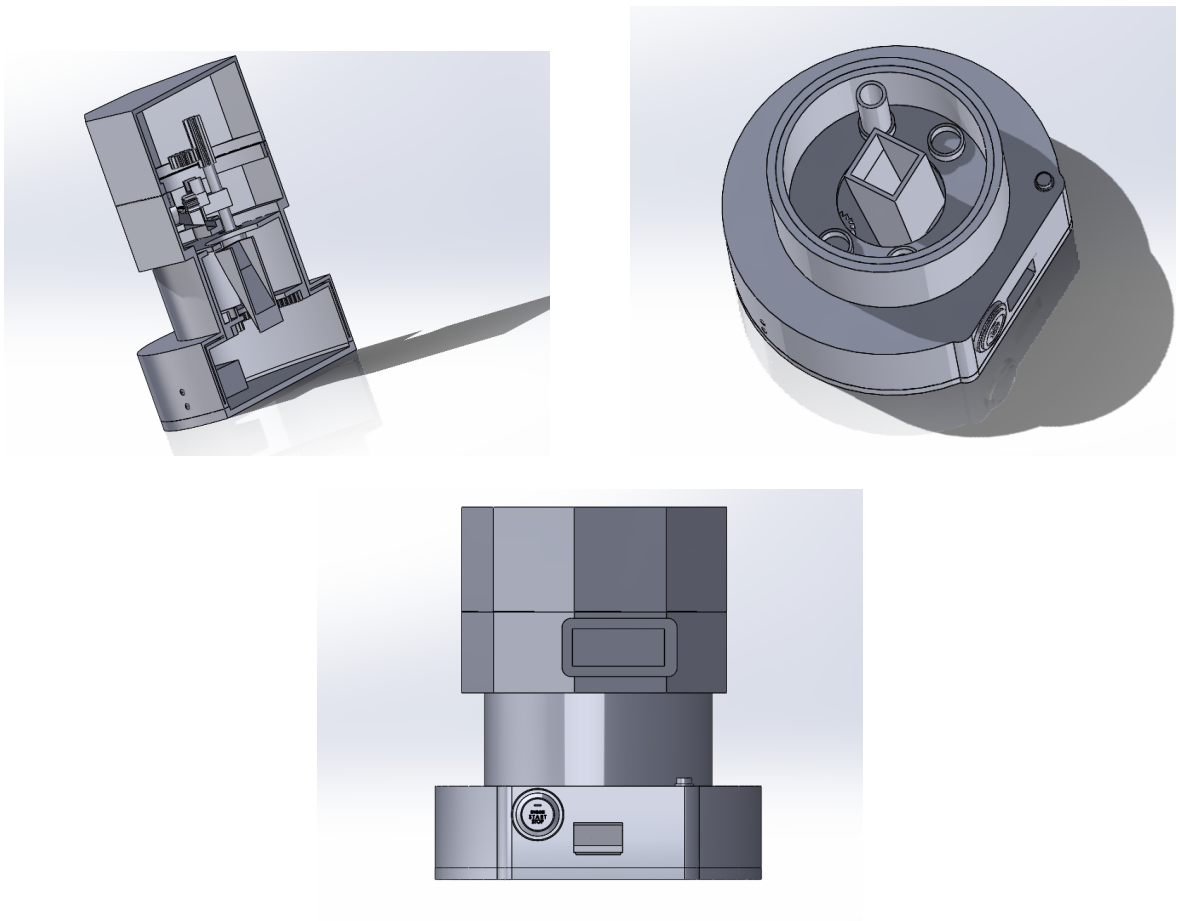


Figure 3.1: Mechanical design of RemindMeBox

Electrical Design

RemindMeBox's electrical system combines core circuitry and control electronics for reliable, safe operation. A central microcontroller manages system logic, scheduling, and actuator control, while dedicated drivers ensure precise pill dispensing.

Infrared and ultrasonic sensors confirm medication retrieval and monitor safety conditions. The user interface includes tactile buttons, an LCD display, and a buzzer for alerts and feedback.

Connectivity via Bluetooth or Wi-Fi enables remote updates and caregiver monitoring. Power is supplied by a stable DC source with a rechargeable backup battery, monitored to prevent failures and alert users when recharging is needed.

All subsystems are isolated and shielded against interference, with fail-safes for sensor, connectivity, and power issues. This design ensures secure, scalable, and reliable system performance.

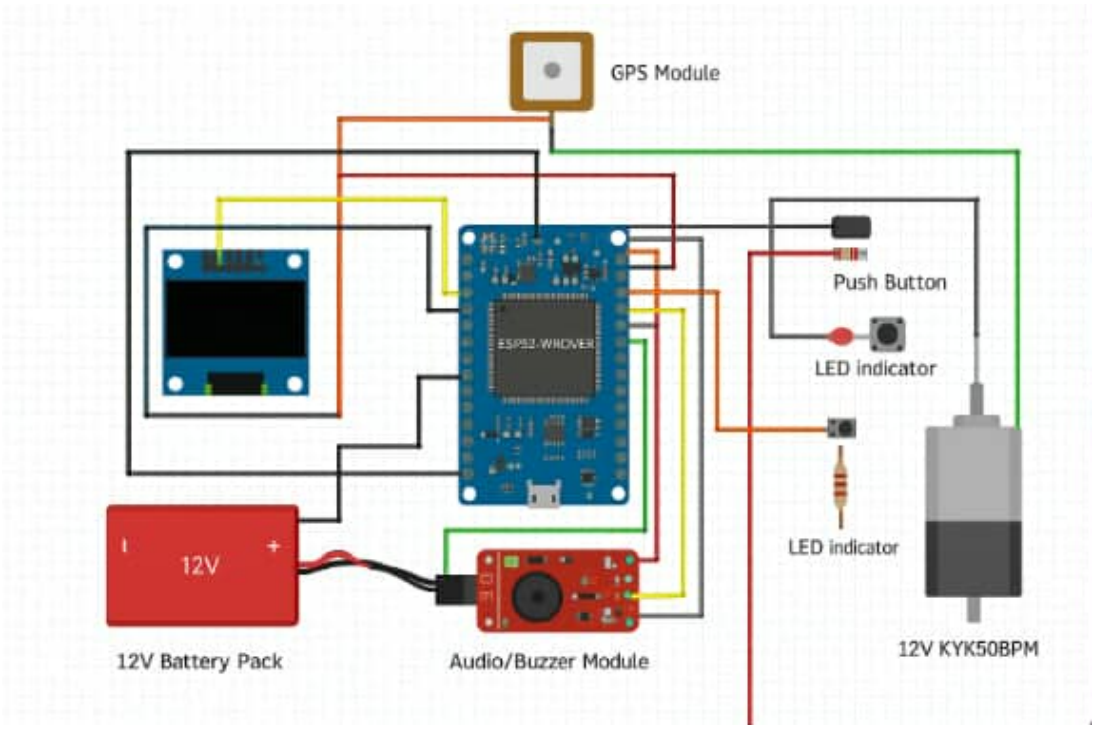


Figure 3.2: RemindMeBox electrical design

3.1.2 Software Architecture

The software architecture of the RemindMe Box ensures secure communication, reliability, and smooth coordination between the ESP32 device, the mobile application, and the

cloud. It is structured around four main components: the ESP32 firmware, the Flutter mobile app, the Firebase cloud backend, and external APIs.

The **ESP32 firmware** manages all real-time operations, including pill dispensing, sensor monitoring, stock updates, and emergency detection. After Bluetooth-based initial setup, it connects to WiFi and synchronizes schedules, compartment data, and alerts with Firebase.

The **mobile application**, built with Flutter, provides authentication, device setup, schedule creation, notifications, mapping features, and user role management (Super User and Simple User). After configuration, the app communicates with the device through Firebase.

The **Firebase backend** centralizes all data. Firestore stores users, devices, schedules, GPS data, and alerts, while Firebase Authentication secures access. Firebase Cloud Messaging (FCM) sends notifications for missed doses, low stock, system errors, or emergencies.

External services enhance the system: the **Gemini API** powers the AI assistant, and **OpenStreetMap/Overpass** provides real-time location and nearby hospital information.

Together, these components deliver a scalable and secure platform for automated medication management, remote monitoring, and caregiver support.

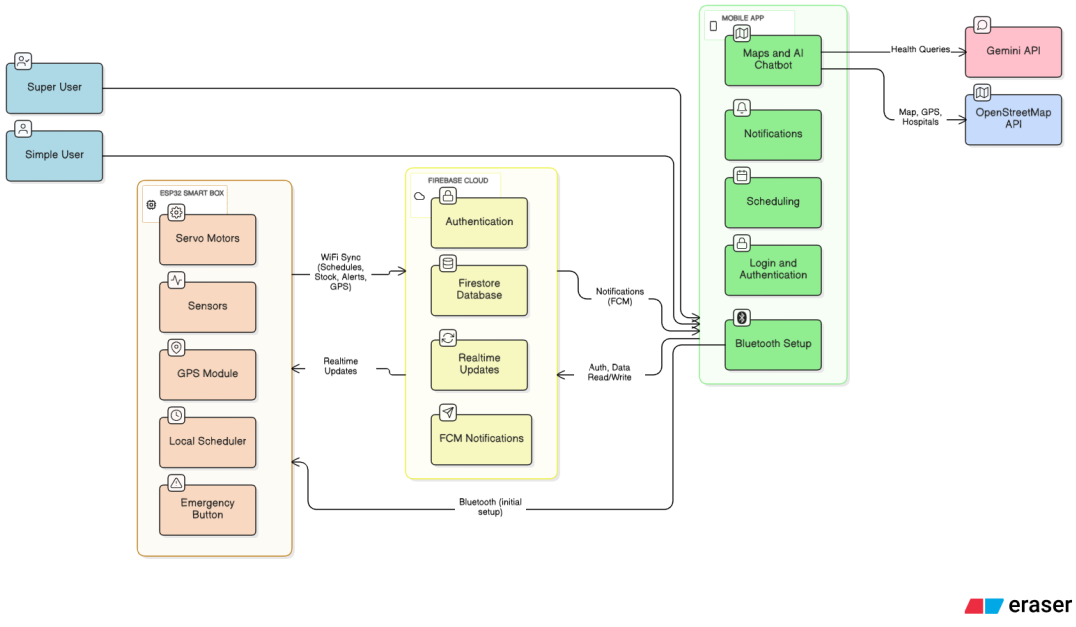


Figure 3.3: Software Architecture Diagram

3.1.3 Data Management & Security

RemindMeBox ensures data integrity, privacy, and reliability through secure local storage and encrypted synchronization using TLS chosen for its strong end-to-end encryption and reliability, protecting all transmitted medical and personal data.

Access to records is role-based for users, caregivers, and healthcare professionals. Secure backups and redundancy prevent data loss, while sharing complies with medical privacy regulations and gives users control over shared information.

All health data is anonymized and encrypted before transmission, with access logs and audit trails maintained. Regular updates address security threats and ensure continued compliance, reinforcing trust and safety across the system.

3.2 Sustainability Considerations

Sustainability is a core design principle of **RemindMeBox**, encompassing environmental, economic, and social aspects. The system is engineered to minimize energy consumption, reduce environmental impact, and maximize accessibility and social benefit for users.

3.2.1 System Overview

RemindMeBox is powered by a 12V rechargeable lithium-ion battery. Power is regulated to 3.3V and 5V through an integrated power management module, ensuring stable operation of all electronic components while optimizing energy efficiency.

3.2.2 Power Management

The device is designed for **long battery life and low power consumption**. Key strategies include the use of energy-efficient components and system sleep modes. Real-time power monitoring alerts users of low battery levels, while a built-in backup supply ensures uninterrupted operation during outages, supporting consistent medication adherence.

3.2.3 Environmental Impact

Sustainability considerations extend to materials and manufacturing. **RemindMeBox** is constructed from recyclable materials and designed for durability and modularity, reducing material usage and facilitating environmentally responsible end-of-life recycling. Manufacturing decisions prioritize materials with a lower environmental footprint, ensuring the device remains sustainable throughout its lifecycle.

3.2.4 Economic and Social Impact

The system emphasizes **affordability and ease of maintenance**, enhancing accessibility for elderly users and patients with chronic conditions. An intuitive interface and remote support minimize the need for frequent technical assistance. By promoting independent living and improving health outcomes, **RemindMeBox** generates significant social value for users, families, and healthcare systems.

Conclusion

RemindMeBox combines reliable hardware, smart software, and secure cloud integration to ensure safe and consistent medication dispensing. Its design prioritizes usability, data security, and sustainability, while supporting remote monitoring and caregiver assistance. Affordable, durable, and energy-efficient, it offers a practical solution that improves medication adherence and health outcomes.

Chapter 4: Implementation and deployment strategy

Introduction

This chapter presents the strategic methodology and implementation plan for **RemindMeBox**, ensuring the system is both **technically robust and secure** and ready for **deployment and adoption** across diverse user environments.

4.1 Development methodology

To deliver an innovative and reliable **smart medication management system**, RemindMeBox adopts the **agile scrum methodology**, a flexible and iterative framework that emphasizes **collaboration, rapid adaptation, and continuous delivery of value**. This methodology aligns perfectly with the project's dynamic environment, where **technical, user, and regulatory feedback** must be quickly integrated to maintain high levels of **usability and safety**. The scrum approach enables the team to **reduce risk, accelerate feedback cycles**, and ensure that the product meets **real-world expectations** through short, focused sprints.

The project team includes dedicated roles reflecting specific expertise. The **product owner** is responsible for defining requirements, gathering user feedback, and prioritizing the backlog. The **scrum master** coordinates sprints, manages workflow, and removes impediments. The **hardware engineer** designs printed circuit boards and integrates sensors and actuators. The **software engineer** develops the mobile application, backend, and artificial intelligence assistant. The **ux/ui designer and tester** ensures interface usability and conducts surveys and testing. Work is organized around project epics and prioritized user stories, with story points illustrating workload and critical features, ensuring that high-priority and high-impact tasks are addressed first.

Sprint 1: Conception and user research

The first sprint focused on **conception and user research**. During this phase, the team conducted user interviews, surveys, and requirements analysis to capture real user pain points. Initial 3D and physical prototype modeling was performed, sprint planning was established, and documentation of design and research outcomes was completed to guide subsequent development.

Sprint 2: Hardware development

The second sprint centered on **hardware development**. Activities included the selection and design of electronic components, sensors, and PCB simulation. The first hardware prototype was assembled and tested, user alert features such as LEDs and buzzers were implemented, and detailed hardware documentation was prepared to ensure reproducibility and reliability.

Sprint 3: Mobile application development

The third sprint concentrated on **mobile application development**. The Flutter-based app interface was designed, and core functionalities including authentication, medication scheduling, reminders, map integration for nearby hospitals, and a chatbot assistant were implemented. All features underwent thorough testing, and comprehensive documentation was produced.

Sprint 4: Integration, testing, and finalization

The fourth and final sprint focused on **integration, testing, and finalization**. Hardware was connected with the mobile application via Bluetooth or Wi-Fi, and full system testing was conducted with real elderly users to ensure reliability. Final documentation and demonstrations were prepared, and the team performed a retrospective to evaluate the process and identify improvements for future iterations.

By structuring development in short, outcome-focused sprints, the scrum framework ensures that RemindMeBox rapidly addresses **true user needs**, adapts to technical challenges, enhances quality and satisfaction, and minimizes development risks at each stage.

The following figures will showcase the Jira plannification of sprints :

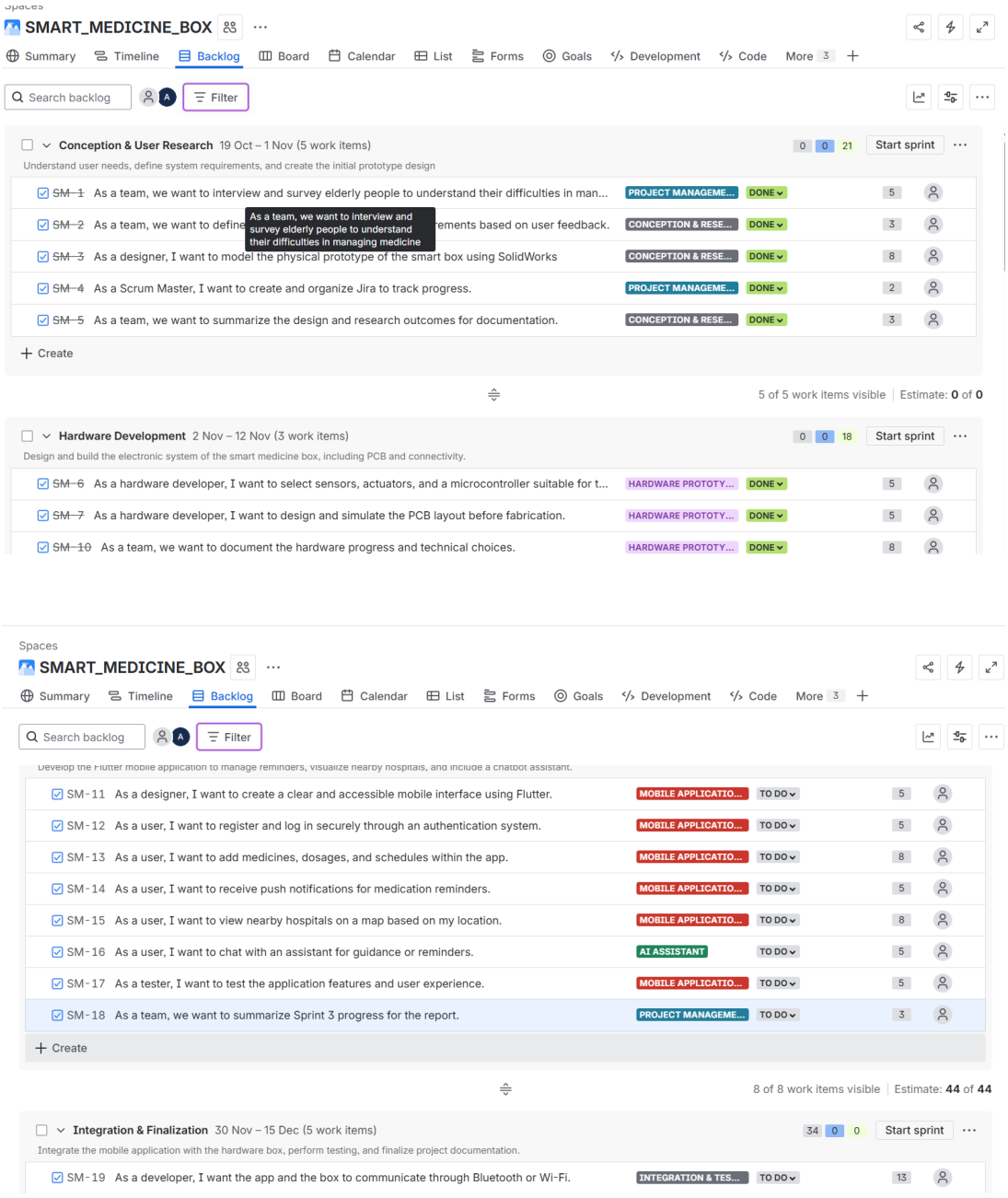


Figure 4.1: Jira planification of sprints

4.2 Market Segmentation and Outreach

To effectively reach its target audience and maximize impact, RemindMeBox adopts a comprehensive market segmentation and strategic partnership approach, aligning its solutions with the specific needs of users, caregivers, healthcare institutions, and community stakeholders .

4.2.1 Market Segmentation

RemindMeBox targets the local smart pill box market, with projected revenues of approximately **100,000 TND** in the first two years.

The system addresses an increasing need for medication adherence solutions, particularly in communities affected by crises, limited healthcare access, or social disruption. The market includes device manufacturing, software services, distribution, and delivery to both individual users and institutional clients.

Key market segments include:

- **Elderly and chronic patients** (35–40%): Require adherence support and continuous monitoring.
- **Caregivers and families** (25%): Focused on safety and convenience.
- **Healthcare institutions** (15–20%): Hospitals, clinics, and pharmacies adopting digital health solutions.
- **NGOs and community programs** (8–12%): Often serving rural or vulnerable populations.
- **Government programs** (5–10%) and other segments such as corporate wellness, insurers, and tech partners .

With strong demand, increasing digital health adoption, and an integrated value chain, RemindMeBox is strategically positioned for growth across both individual and institutional healthcare segments, particularly in regions where access to continuous care is disrupted or insufficient.

4.2.2 Strategic Partners

Strategic partners provide RemindMeBox with essential expertise, distribution, and credibility enabling faster adoption and broader impact across healthcare and community networks.

Government	Private Sector	International Orgs	Academia	NGOs	Community
Ministry of Health Tunisia	Telnet, Pharmaghreb	WHO	National Universities of Tunisia	Associamed Tunisia	Dar Ellama Retirement House
CNAM	Novation City	World Bank	Institut Pasteur de Tunis	RedStart Tunisia	Beyt El Izz Retirement House
Ministry of Social Affairs	Flat6Labs Tunis	UNICEF	National Engineering Schools	Lab'ess	Local patient advocacy groups

Table 4.1: Key Strategic Partners for RemindMeBox

4.3 Business model canvas

The Business Model Canvas of RemindMeBox illustrates how the solution creates value, connects with users, and operates sustainably.

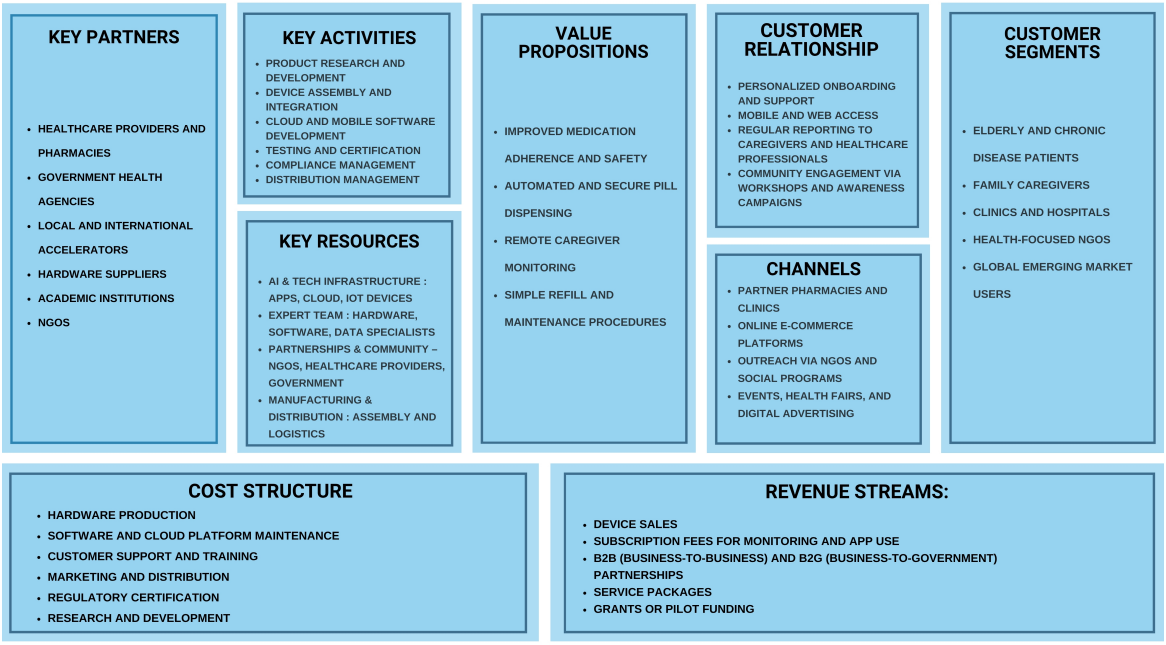


Figure 4.2: RemindMeBox's Business Model Canvas

4.4 Competitors analysis and market differentiators

RemindMeBox stands out with its lower production cost and tailored features for the local market, while competitors are often pricier or lack integrated physical and digital capabilities.

4.4.1 Direct and indirect competitors

RemindMeBox faces direct competition from global smart dispensers and emerging local startups, as well as indirect competition from simple pill boxes and reminder apps

Competitor/Product	**Type**	**Region**	**Key features**
MedMinder	Direct	International	Wireless cloud connection, remote monitoring, alerts for missed doses
Lifeline Medready	Direct	International	Simple design, affordable, SMS alerts, monitored by call center
Med-E-Lert	Direct	International	Programmable alerts, simple design, battery operated

Competitor/Product	Type	Region	Key features
Local pharmacy pill boxes	Indirect	Tunisia	Manual pill organizers, no connectivity, low cost
Smartphone reminder apps (Medisafe, Google Calendar)	Indirect	Tunisia	Medication reminders only, not physical dispensing
Pharmaghreb HealthTech devices (future local)	Direct	Tunisia	Emerging connected medication solutions

Table 4.2: Direct And Indirect Competitors

4.4.2 Factors Boosting RemindMeBox Competitiveness

RemindMeBox gains a competitive edge through several key factors. Its adaptable design allows the system to meet the needs of healthcare environments disrupted by crises, accommodating local languages, practices, and cultural contexts.**Affordability** is achieved through cost-effective device and service packages suitable for emerging markets. The platform offers **integrated features**, including physical dispensing, cloud monitoring, multi-user notifications, and mobile and web access. A robust **partnership ecosystem** with local NGOs, health agencies, and pharmacies enhances reach and adoption. **Regulatory alignment** is ensured through early compliance with national health and medical device standards. The system's **easy refill and support** design simplifies maintenance and provides comprehensive training for both users and caregivers. **Data privacy and security** are prioritized with end-to-end encryption, strong authentication, and compliance with GDPR and local regulations. **Pilot programs** have validated its effectiveness through national deployments, and the solution demonstrates strong **scalability potential**, ready for expansion across MENA and francophone Africa via local partnerships.

4.5 Risk Management

To ensure its safety, reliability, and effectiveness , targeted mitigation strategies have been implemented to address potential operational and user-related risks

4.5.1 SWOT Analysis

To understand the internal strengths and weaknesses , as well as external opportunities and threats, a SWOT analysis was conducted.

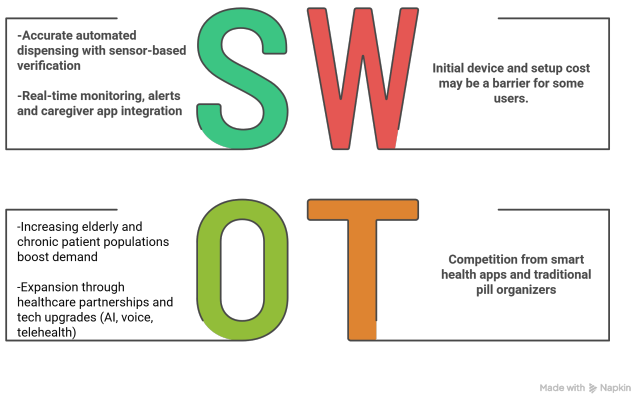


Figure 4.3: SWOT analysis

4.5.2 PESTEL Analysis

To evaluate the broader macro-environmental factors that could impact RemindMeBox's success, a PESTEL analysis was performed, highlighting political, economic, social, technological, environmental, and legal influences.

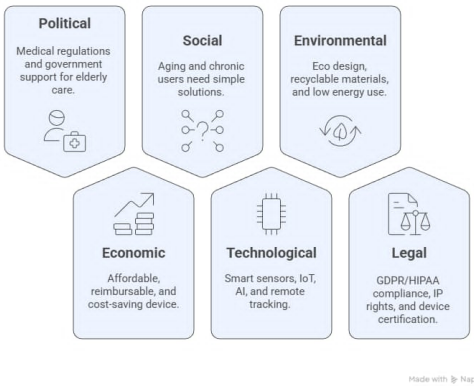


Figure 4.4: PESTEL analysis

4.5.3 Risk Mitigation

Based on the insights from SWOT and PESTEL analyses, specific strategies were designed to minimize operational and user-related risks. Missed dose or user refusal is addressed by automatically recording events and sending alerts to caregivers, enabling timely follow-up and reducing prolonged non-adherence. Network or power disruptions, which are common in regions experiencing overlapping crises such as conflict, natural disasters, or infrastructure breakdowns, are mitigated by local storage of all scheduled tasks and operational records, with automatic synchronization once systems are restored. Overdose risk is minimized by preventing multiple dispensing attempts within restricted periods, and caregivers are immediately notified if abnormal dispensing behavior is detected, ensuring rapid intervention and user safety.

Conclusion

RemindMeBox combines resilient hardware, intuitive software, and secure cloud integration to ensure medication adherence even in regions affected by crises. Its agile development approach allows rapid adaptation to user needs and technical challenges in unstable environments. Strategic partnerships with NGOs, health agencies, and local providers, along with built-in risk mitigation for network or power disruptions, ensure reliability, safety, and scalability. Overall, it delivers a practical, affordable, and impactful solution that maintains continuity of care for patients, caregivers, and healthcare systems under challenging conditions.

A Supplementary Diagrams

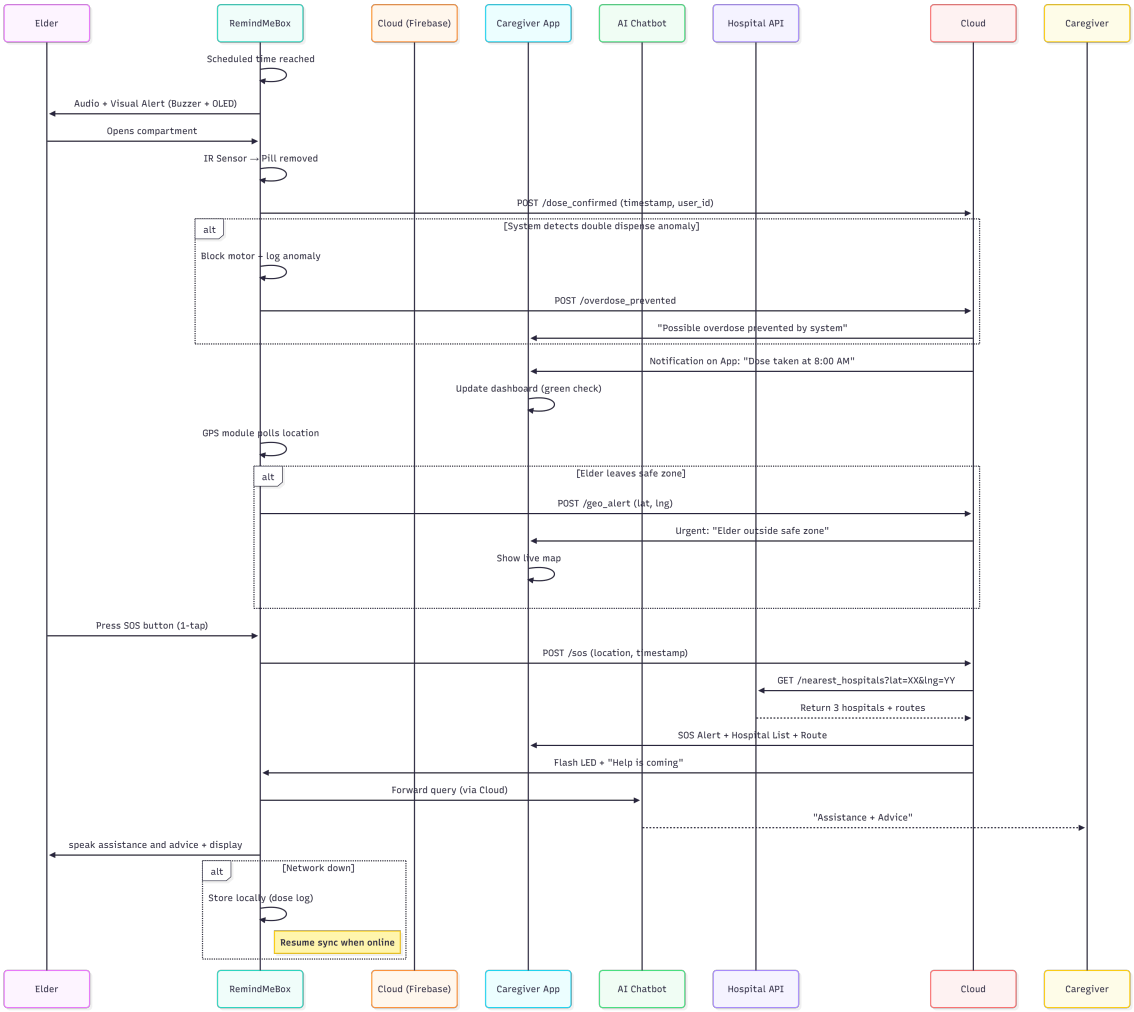


Figure A.1: Sequence Diagram of RemindMeBox Function